

Course: Wireless and Mobile Security (CY315)
Semester: Spring-2026
Instructor: Dr. Muhammad Fawad Khan, AP

Project Overview Guidelines

Research-Based Project and Optimization Tracks

1. Project Overview

Modern wireless networks (5G/6G, IoT, Wi-Fi 6) face sophisticated threats at both the protocol and physical layers. Standard security often fails against intelligent jamming, cross-layer attacks, or resource-constrained environments.

Students are required to conduct a research-based project that either critically evaluates existing WMS solutions (Survey Track) or optimizes/enhances a specific security mechanism (Implementation Track). The goal is to move beyond basic concepts to address:

- **Robustness:** Protecting communication against active eavesdropping or interference.
- **Efficiency:** Optimizing security overhead for mobile/constrained devices.
- **Contemporary Solutions:** Applying Physical Layer Security (PLS) or advanced cryptography to modern wireless problems.

2. Project Tracks

Track A: Implementation & Optimization Track

For students choosing the implementation track, the goal is to implement, evaluate, or optimize a specific wireless security mechanism. Projects can focus on physical layer security, jammer mitigation, key exchange protocols in mobile networks, or lightweight authentication systems. Deliverables must include fully documented simulation or implementation code (e.g., in Python, NS-3, or MATLAB) and a comparative performance report.

Track B: Research Survey & Literature Review Track

For students choosing the survey track, the goal is to perform a rigorous, systematic literature review on a contemporary wireless security paradigm. Your review must identify at least 15-20 high-quality recent papers, categorize them into thematic taxonomies, critically identify research gaps, and propose solid future directions. The resulting paper must follow the IEEE conference formatting guidelines.

3. Deliverables & Milestones

Milestone	Deadline	Description
Project Proposal	Week 6	1-2 page proposal outlining track, problem statement, and initial references.
Progress Report	Week 11	Mid-term report showing completed model/survey layout and initial results.
Final Submission	Week 16	Final research paper (IEEE PDF format), slide deck, and source code (if Track A).

4. Grading Criteria

- **Problem Definition & Technical Depth:** 40% (Track A: Implementation correctness and performance; Track B: Analysis of state-of-the-art and taxonomy).
- **Documentation & IEEE Formatting:** 30% (Quality of the final report, structure, and academic referencing).
- **Presentation & Defense:** 30% (Defense of the methodology, design choices, and results in front of the project panel).